

Reason about equivalent equations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution? Copy or complete the first step.

2. Solve $x + 38 = 91$.

3. Solve $5x - 7 = 48$.

4. Miro says: Miro changes one side of an equation without making the same balanced change to the other side. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution? Copy or complete the first step.
Answer Yes, both give $x = 7$.
2. Solve $x + 38 = 91$.
Answer $x = 53$.
3. Solve $5x - 7 = 48$.
Answer $x = 11$.
4. Miro says: Miro changes one side of an equation without making the same balanced change to the other side. Is this correct? Explain.
Answer Use an inverse operation on the whole equation and check by substitution.

Reason about equivalent equations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

6. Check this result using an inverse, estimate or known fact:
Solve $5x - 7 = 48$.

2. Solve $x + 38 = 91$.

7. Prove or explain your reasoning: Solve $6(x + 4) = 72$.

3. Solve $5x - 7 = 48$.

8. Identify and correct this misconception: Miro changes one side of an equation without making the same balanced change to the other side.

4. Solve this independently and show every stage: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Solve $x + 38 = 91$.

10. Write and solve a similar question to: Solve $5x - 7 = 48$.

Teacher answers and checking prompts

1. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Yes, both give $x = 7$.
2. Solve $x + 38 = 91$.
Answer $x = 53$.
3. Solve $5x - 7 = 48$.
Answer $x = 11$.
4. Solve this independently and show every stage: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Yes, both give $x = 7$.
5. Use a second method or representation for: Solve $x + 38 = 91$.
Answer Expected result: $x = 53$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Solve $5x - 7 = 48$.
Answer Expected result: $x = 11$. A valid check must be shown.
7. Prove or explain your reasoning: Solve $6(x + 4) = 72$.
Answer $x = 8$. The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes one side of an equation without making the same balanced change to the other side.
Answer Use an inverse operation on the whole equation and check by substitution.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Solve $5x - 7 = 48$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason about equivalent equations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Solve $6(x + 4) = 72$.

6. Create a new example that tests the same idea as: Solve $5x - 7 = 48$.

2. Prove the answer to this question, rather than only calculating it: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Solve $x + 38 = 91$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $x = 11$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes one side of an equation without making the same balanced change to the other side.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Solve $6(x + 4) = 72$.
Answer $x = 8$. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Expected result: Yes, both give $x = 7$. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Solve $x + 38 = 91$.
Answer Expected result: $x = 53$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $x = 11$.
Answer One valid reconstruction is: Solve $5x - 7 = 48$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes one side of an equation without making the same balanced change to the other side.
Answer Use an inverse operation on the whole equation and check by substitution.
6. Create a new example that tests the same idea as: Solve $5x - 7 = 48$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.